

Quant researcher, econometrician, economist and data scientist

Profile

Accomplished quantitative researcher, econometrician, economist and data scientist with 10 years of experience in central banking and the finance industry.

Professional experience

- From May 2021 to February 2026

● **Quantitative researcher**
Qube research and Technologies Paris, France

 - Developed macroeconomic models (Bayesian VARs, Phillips curves, Taylor rules, Nelson-Siegel) and machine learning models (neural networks, XGBoost and random forests) to generate global macro signals and implement investment strategies.
 - Designed and coded nowcasting models (dynamic factor models) to predict macroeconomic conditions and generate macroeconomic alpha on a daily basis.
 - Created a full Database of point-in-time macroeconomic indicators based on Datastream.
 - Enhanced classical investment strategies with macro indicators, e.g. macro trends for momentum, real activity indicators for FX carry, or real business cycles for equity value.
- From September 2020 to April 2021

● **Quantitative researcher**
Capital Fund Management Paris

 - Developed and coded a wide range of predictive models for the main macroeconomic aggregates: econometrics (VAR, Bayesian VAR, time-varying Bayesian VAR), nowcasting (dynamic factor model, MIDAS regression, mixed frequency Bayesian VAR), and machine learning (LSTM, random forest and gradient boosting).
 - Realised a systematic comparison of the predictive and directional performances of the different models.
- From September 2014 to December 2018

● **Econometrician and economist**
European Central bank Frankfurt-am-Main, Germany

 - Developed the Bayesian Estimation, Analysis and Regression (BEAR) Toolbox, an advanced Bayesian time-series software for forecasting and economic/financial analysis.
More about BEAR: <https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1934.en.pdf>
 - Organised and animated research seminars in econometrics. Contributed to scientific articles. Trained economists to Bayesian econometric methods.
 - Drafted economic articles and policy notes.

Education

- From 2019 to 2020

● **Post-Master's degree in Big data**
Télécom ParisTech Paris
- From 2008 to 2014

● **PhD in Economics, with a specialization in econometrics and time-series methods**
University of Cergy Pontoise - ESSEC Business School Cergy
- From 2007 to 2008

● **Research Master in Economics**
University of Cergy-Pontoise / ESSEC Business School Cergy
- From 2003 to 2007

● **Sciences Po**
Institute of Political Studies Lyon
- From 2001 to 2004

● **Bachelor in Foreign Languages, English and Chinese**
University Jean Moulin Lyon 3 Lyon

Publications and Working papers

- Time-varying Vector Autoregressions: efficient estimations, random inertia and random mean. *MPRA working paper*, August 2019.
- The Bayesian Estimation, Analysis and Regression (BEAR) toolbox – with Alistair Dieppe and Björn Van Roye. *ECB Working Paper Series No. 1934*, July 2016.
- Euro introduction: has there been a structural change? Study on 10 - European Union countries. *Economic Modelling No. 40*, June 2014.
- L'effet dynamique des chocs d'offre et de demande agrégés. Une étude sur le cas allemand. (The dynamic effect of aggregate supply and aggregate demand shocks: study on the German case). *Revue Economique No. 63*, January 2012.

Personal projects

Alexandria: Alexandria is a new software/library for Bayesian econometrics and time-series models. It is free and double-coded in Matlab and Python. It comes with a series of free books that provide full theory and algebraic derivations behind the models. Alexandria is being constantly updated and includes the following models:

- Bayesian linear regression
- Bayesian vector autoregression, with multiple priors and wide range of state-of-the art applications (predictions, impulse response function, variance and historical decomposition, conditional forecasts, structural identification by zero, sign and narrative restrictions).
- Bayesian vector error correction (BVEC) and Bayesian vector autoregressive moving average (BVARMA)
- Bayesian nowcasting: mixed frequency Bayesian VAR, Bayesian dynamic factor model, and Bayesian MIDAS regression. All models allowing for mixed frequency and missing observations

Video presentation of the Bayesian VAR models: <https://www.youtube.com/watch?v=SXfmmfSZkOc&t=145s>

Alexandria website, with all downloads: <https://lnkd.in/euKQC3hj>

Alexandria Github repo, with all downloads: <https://lnkd.in/ezqUinWF>

Technical skills

High proficiency:

- Python - Numpy, Pandas, Matplotlib, ScikitLearn, Keras, Tensorflow
- Matlab - Dynare
- Github / Gitlab
- Office Suite

Intermediate proficiency:

- Sql-NoSql (PostgreSQL, Cassandra)
- distributed frameworks (Spark, Hadoop)
- Java programming

Personal interests

Mixed Martial Arts

Swimming

Travelling

Languages

French: native

English: fluent